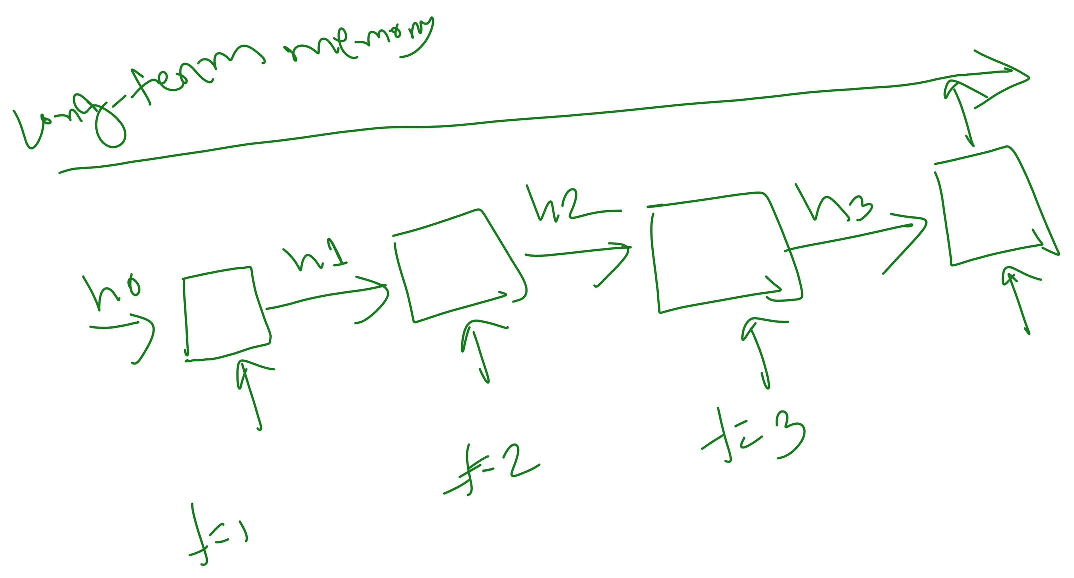
LSTM Architecture

1) Long term memory   2) Short-term

Simple overview



- **Short-Term Memory** → what is happening right now
- **Long-Term Memory** → important information saved for later

🎯 Story Example: "The Lost Puppy"

Teacher says this story word by word:

"Yesterday, **Rahim** lost his puppy in the park. , Rahim

Long-term memory = Rahim

Short term = he
3 hours
lake

He searched everywhere.

After 3 hours, **he** finally found it near a lake."

Now ask students:

👉 Who found the puppy?

They answer: **Rahim**

🧠 How LSTM Thinks

1️⃣ Long-Term Memory (Important Information)

When the story first says:

"Rahim lost his puppy…"

LSTM thinks:

✅ "Rahim is important. I should remember this for later."

So it stores **Rahim** in the **long-term memory cell**.

2️⃣ Short-Term Memory (Current Information)

While reading:

"searched everywhere…"

"after 3 hours…"
"near a lake…"

These are temporary current details.

LSTM keeps these in **short-term memory** because they help understand the current sentence.

But later many of these details may be forgotten.

🔥 The Important Moment

When the model reads:

"he finally found it…"

LSTM asks:

👉 "Who is *he*?"

It checks the **long-term memory** and finds:

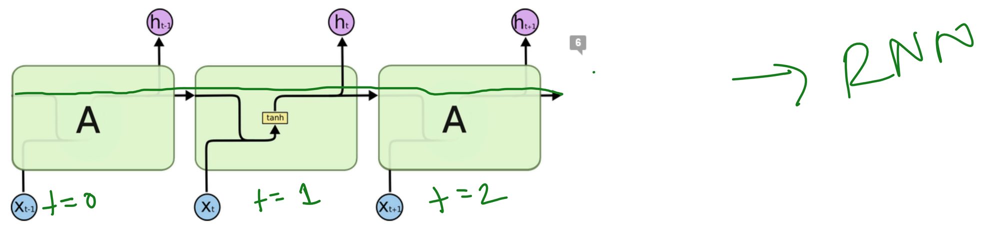
✅ "Rahim"

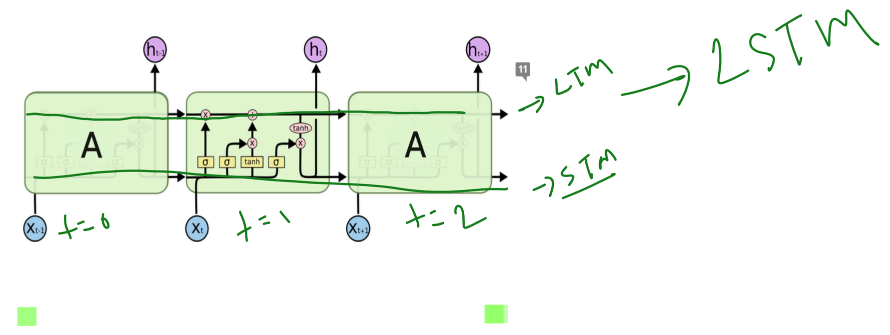
So the model correctly understands:

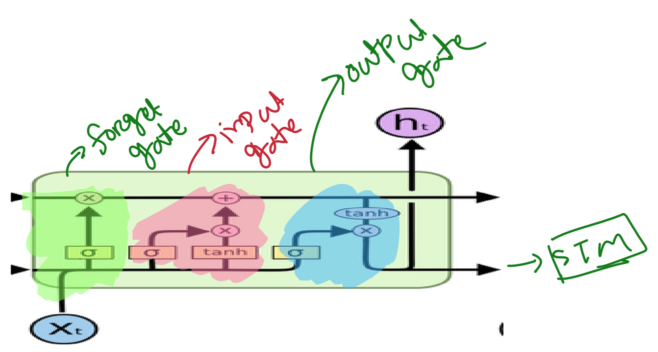
Rahim found the puppy.

🎓 Analogy

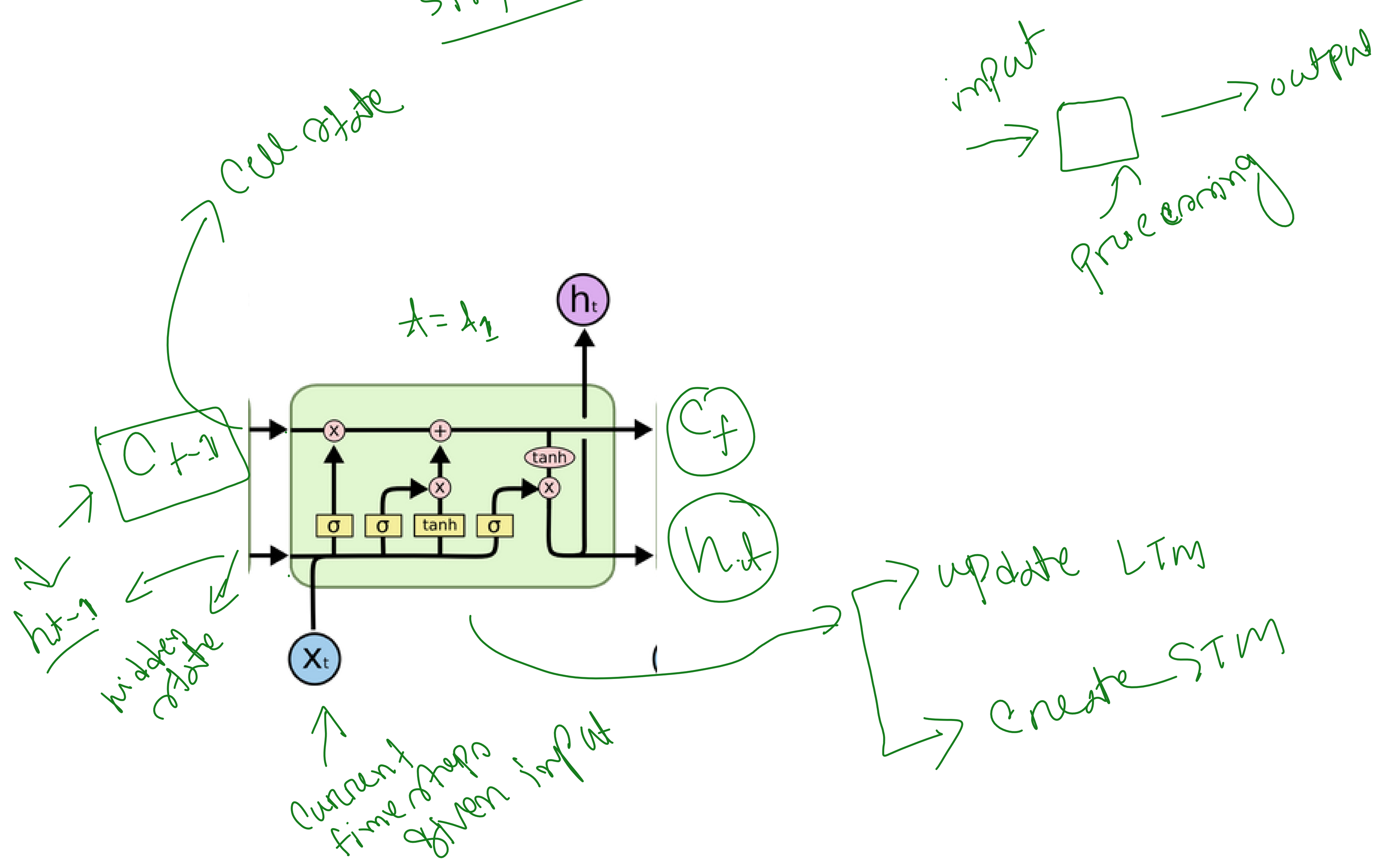
Human Brain	LSTM
Important life memories	Long-Term Memory
What you are thinking right now	Short-Term Memory
Forgetting unimportant details	Forget Gate
Saving important information	Input Gate
Using remembered info later	Output Gate







Simple Analysis



🔴 1. Forget Gate

~~The **Forget Gate** decides:~~

"What old information is no longer important?"

🎯 Real-Life Example

Story:

"Rahim was living in Dhaka.

Later he moved to Chittagong."

Now the old location (**Dhaka**) is no longer important.

The LSTM says:

❌ Forget "Dhaka"

✅ Keep "Chittagong"

🧠 What It Does

It looks at:

- Previous long-term memory
- Current input

Then decides:

- Keep information
- Or erase information

🟢 2. Input Gate

The **Input Gate** decides:

“What NEW information should I save?”

🎯 Example

Story:

“Rahim got a new puppy.”

LSTM thinks:

✅ “New puppy is important.”

So it stores this in long-term memory.

🔵 3. Output Gate

The **Output Gate** decides:

“What information should I use RIGHT NOW?”

🎯 Example

Sentence:

“Rahim lost his puppy…

After hours, he found it.”

To understand “he”:

LSTM uses stored memory:

✅ “Rahim”

The output gate exposes this useful information.

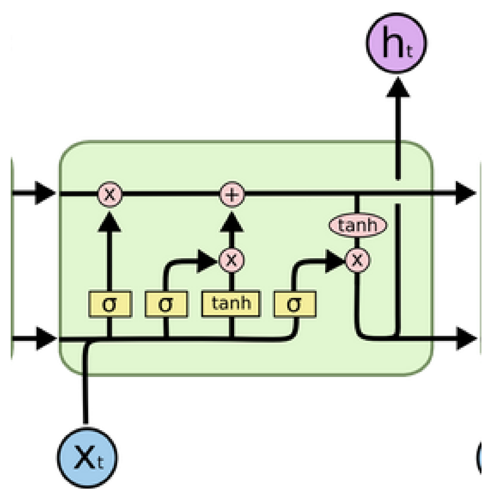
🧠 What It Does

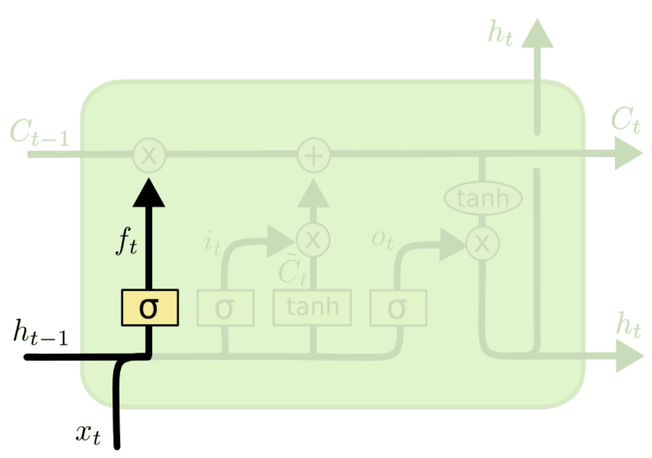
It chooses:

- Which memory is relevant NOW
- What should become short-term memory/output

🎓 Simple One-Line Definitions

Gate	Job
Forget Gate	Remove useless information
Input Gate	Save important new information
Output Gate	Use relevant information now





$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] \; + \; b_f\right)$$